

KANIN NOS, Russian Federation

WMO: 221650

Lat: 68.655N

Lon: 43.288E

Elev: 49

StdP: 100.74

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.3	-19.8	-25.3	0.4	-23.0	-21.8	0.5	-19.5	20.9	-9.2	19.4	-9.9	7.1	90	1.024

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	18.2	14.1	16.0	13.2	14.1	12.0	14.8	17.3	13.5	15.6	12.2	13.8	6.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	9.7	16.3	12.3	8.9	14.6	11.2	8.3	13.2	41.8	17.3	38.1	15.7	34.8	13.9	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 17.9	15.8	14.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-23.3	16.1	3.6	2.7	-25.9	18.0	-28.0	19.6	-30.1	21.1	-32.7	23.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-8.1	-9.0	-7.1	-3.3	0.2	5.4	9.0	8.9	7.0	2.4	-1.7	-4.4	(6)
(7)		DBStd	7.44	5.54	6.21	4.64	3.30	3.41	3.74	3.77	2.57	2.22	2.70	3.61	4.63	(7)
(8)		HDD10.0	3719	560	531	530	400	303	148	67	54	94	235	350	447	(8)
(9)		HDD18.3	6693	818	764	788	650	561	387	291	293	341	493	600	706	(9)
(10)		CDD10.0	69	0	0	0	0	1	11	35	19	3	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	8.2	8.6	7.5	6.9	6.7	6.4	6.3	6.2	6.8	7.6	8.0	8.6	(14)
(15)	Precipitation	PrecAvg	426	35	28	25	22	23	33	38	50	46	52	40	37	(15)
(16)		PrecMax	661	118	83	87	56	86	112	100	119	125	102	74	86	(16)
(17)		PrecMin	257	5	1	4	7	4	5	4	3	12	16	12	10	(17)
(18)		PrecStd	76	21	16	15	10	13	20	24	28	20	18	14	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	0.7	0.8	4.3	12.6	19.6	22.7	18.1	14.1	8.6	5.6	2.9	(19)
(20)			MCWB	0.4	0.3	-0.1	2.1	9.0	14.4	16.9	14.7	11.0	7.7	5.1	2.2	(20)
(21)		2%	DB	0.1	-0.4	0.1	2.2	9.1	16.5	19.2	15.9	11.8	7.9	4.4	1.9	(21)
(22)			MCWB	-0.5	-0.9	-0.5	1.0	6.6	12.7	14.7	13.7	10.7	7.2	3.8	1.2	(22)
(23)		5%	DB	-0.6	-1.2	-0.5	0.9	6.6	14.1	17.1	14.0	10.8	7.1	3.4	1.1	(23)
(24)			MCWB	-1.1	-1.6	-1.0	0.3	5.0	11.1	13.9	12.7	9.9	6.5	2.8	0.4	(24)
(25)		10%	DB	-1.3	-2.1	-1.2	0.3	4.7	11.6	15.1	12.6	10.0	6.2	2.6	0.4	(25)
(26)			MCWB	-1.8	-2.5	-1.6	-0.2	3.5	9.2	12.9	11.4	9.3	5.5	2.0	-0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.4	0.2	2.3	9.2	15.5	17.5	15.3	11.9	8.0	5.4	2.4	(27)
(28)			MCDWB	0.8	0.4	0.5	3.8	12.2	18.3	21.1	17.3	12.8	8.3	5.5	2.7	(28)
(29)		2%	WB	-0.4	-0.7	-0.4	1.1	6.7	13.0	15.6	13.8	10.8	7.3	3.9	1.3	(29)
(30)			MCDWB	-0.1	-0.5	-0.1	1.9	8.5	15.9	18.5	15.6	11.6	7.7	4.2	1.7	(30)
(31)		5%	WB	-1.0	-1.5	-0.9	0.5	5.0	11.0	14.2	12.7	10.1	6.5	2.9	0.6	(31)
(32)			MCDWB	-0.7	-1.2	-0.5	1.0	6.4	13.8	16.7	14.0	10.8	7.0	3.4	0.9	(32)
(33)		10%	WB	-1.8	-2.5	-1.5	0.0	3.6	9.1	12.9	11.7	9.4	5.7	2.1	-0.1	(33)
(34)			MCDWB	-1.4	-2.1	-1.1	0.3	4.6	11.6	15.0	12.6	10.0	6.3	2.5	0.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	4.1	4.0	3.3	3.3	4.1	4.1	3.5	2.5	2.4	3.1	3.5	(35)
(36)			MCDBR	3.1	3.1	3.2	3.4	6.6	8.7	6.9	6.1	3.6	2.6	2.8	2.8	(36)
(37)			MCWBR	3.1	3.2	2.8	3.0	5.1	6.1	4.5	4.4	2.9	2.7	2.8	2.9	(37)
(38)		5% WB	MCDBR	3.2	3.1	2.8	3.2	6.4	8.5	6.5	5.6	3.5	2.5	2.7	2.9	(38)
(39)			MCWBR	3.2	3.3	2.7	2.9	5.1	6.3	4.7	4.3	2.9	2.7	2.9	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.375	0.269	0.307	0.334	0.339	0.336	0.366	0.345	0.321	0.289	0.361	N/A	(40)	
(41)		taud	2.570	2.066	2.151	2.250	2.416	2.518	2.457	2.502	2.504	2.220	2.540	N/A	(41)	
(42)		Ebn at Noon	82	483	703	805	852	867	820	799	721	464	66	0	(42)	
(43)		Edh at Noon	12	47	80	98	94	88	91	78	59	40	10	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.02	0.32	1.59	3.16	4.33	5.14	4.65	3.06	1.67	0.47	0.06	0.00	(44)	
(45)		RadStd	0.00	0.07	0.18	0.27	0.33	0.61	0.56	0.33	0.20	0.07	0.01	0.00	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page